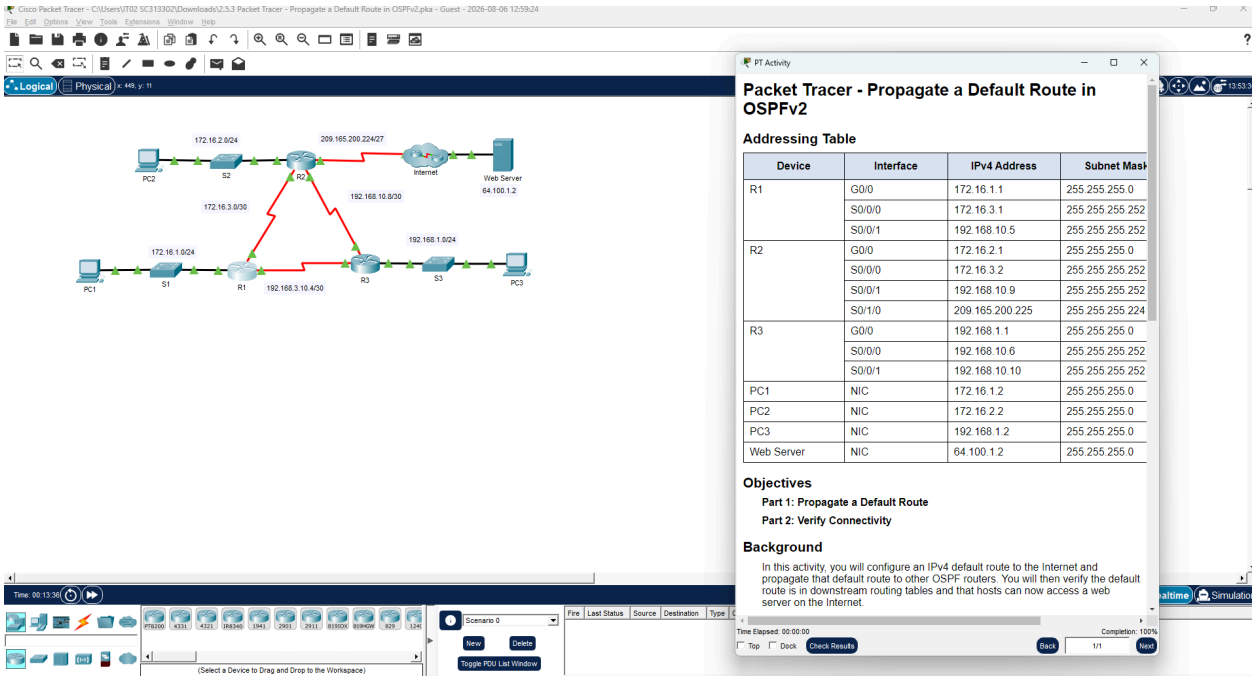


Topology

R1, R2, and R3 form the internal OSPFv2 routing domain (Area 0). R2 additionally connects out to the Internet (and the Web Server) via interface Serial0/1/0, a link that sits outside the OSPF autonomous system. Because R1 and R3 have no direct route to the Internet, R2 must be configured with a default route and must redistribute (propagate) that default route into OSPF so R1 and R3 learn it automatically.

Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	G0/0	172.16.1.1	255.255.255.0	N/A
R1	S0/0/0	172.16.3.1	255.255.255.252	N/A
R1	S0/0/1	192.168.10.5	255.255.255.252	N/A
R2	G0/0	172.16.2.1	255.255.255.0	N/A
R2	S0/0/0	172.16.3.2	255.255.255.252	N/A
R2	S0/0/1	192.168.10.9	255.255.255.252	N/A
R2	S0/1/0	209.165.200.225	255.255.255.224	N/A
R3	G0/0	192.168.1.1	255.255.255.0	N/A
R3	S0/0/0	192.168.10.6	255.255.255.252	N/A
R3	S0/0/1	192.168.10.10	255.255.255.252	N/A
PC1	NIC	172.16.1.2	255.255.255.0	172.16.1.1
PC2	NIC	172.16.2.2	255.255.255.0	172.16.2.1

PC3	NIC	192.168.1.2	255.255.255.0	192.168.1.1
Web Server	NIC	64.100.1.2	255.255.255.0	64.100.1.1

Objectives

Part 1: Propagate a Default Route

Part 2: Verify Connectivity

Background

In this activity, an IPv4 default route to the Internet is configured on R2 and propagated to the other OSPF routers (R1 and R3). The default route is then verified in the downstream routing tables, and connectivity from PC1, PC2, and PC3 to the Web Server on the Internet is confirmed.

Part 1: Propagate a Default Route

Step 1: Test connectivity to the Web Server

From PC1, PC2, and PC3, ping the Web Server at 64.100.1.2.

Were any of the pings successful?

Answer: No. All pings from PC1, PC2, and PC3 to the Web Server (64.100.1.2) fail.

What message did you receive, and which device issued the message?

Answer: "Destination host unreachable" (U.U.U marks in the ping output) - the message is issued by R2, the last-hop router before the Internet, because it has no route to the 64.100.1.0/24 network.

Step 2: Examine the routing tables on R1, R2, and R3

The output of "show ip route" on each router was examined.

What statement is present in the routing tables that indicates that the pings to the Web Server will fail?

Answer: "Gateway of last resort is not set" appears in the routing table of each router. Without a gateway of last resort (a default route), a router drops any packet destined for a network that is not explicitly listed in its routing table - which is the case for the Web Server's 64.100.1.0/24 network.

Step 3: Configure a default route on R2

R2 is configured with a static default route pointing out its Internet-facing interface, Serial0/1/0:

```
R2(config)# ip route 0.0.0.0 0.0.0.0 Serial0/1/0
```

Note: The router issues a warning that using an interface (rather than a next-hop IP address) as the exit point may impact performance if the link is not point-to-point. This warning can be safely ignored here because Serial0/1/0 is a point-to-point serial connection.

Step 4: Propagate the default route in OSPF

OSPF is configured to advertise (originate) the default route to its neighbors:

```
R2(config)# router ospf 1
 R2(config-router)# default-information originate
```

Step 5: Examine the routing tables on R1 and R3

After the default route is propagated, R1 and R3 each learn the route as an OSPF External Type 2 (O*E2) route:

```
R1> show ip route
 <output omitted>
 Gateway of last resort is 172.16.3.2 to network 0.0.0.0
 <output omitted>
 O*E2  0.0.0.0/0 [110/1] via 172.16.3.2, 00:00:08, Serial0/0/0

R3> show ip route
 <output omitted>
 Gateway of last resort is 192.168.10.9 to network 0.0.0.0
 <output omitted>
 O*E2  0.0.0.0/0 [110/1] via 192.168.10.9, 00:08:15, Serial0/0/1
```

R1 learns the default route from R2 via the direct R1-R2 serial link (172.16.3.0/30, out Serial0/0/0). R3 learns the default route from R2 via the R2-R3 serial link (192.168.10.8/30, out Serial0/0/1) since that is R3's lowest-cost/next-hop path to R2 through the triangle of serial links.

Part 2: Verify Connectivity

With the default route now propagated throughout the OSPF domain, PC1, PC2, and PC3 can each successfully ping the Web Server (64.100.1.2).

Can PC1 ping the Web Server?

Answer: Yes. Reply from 64.100.1.2: bytes=32 time<1ms TTL=125 (4/4 successful pings).

Can PC2 ping the Web Server?

Answer: Yes. Reply from 64.100.1.2: bytes=32 time<1ms TTL=125 (4/4 successful pings).

Can PC3 ping the Web Server?

Answer: Yes. Reply from 64.100.1.2: bytes=32 time<1ms TTL=125 (4/4 successful pings).

Why did the pings succeed after configuring R2?

Answer: Once R2 had both a default route (0.0.0.0/0 via S0/1/0) and "default-information originate" enabled under OSPF, R1 and R3 received that default route as an O*E2 entry and set their own gateway of last resort. Every router along the path now has a matching route for the Web Server's network (either specific or default), so packets are no longer dropped.

Final Router Configurations

Full running configurations for R1, R2, and R3 after completing the activity are shown below. (Baseline settings such as hostnames, interface addressing, and OSPF area 0 adjacencies were already in place from earlier labs in this series; the default route and "default-information originate" on R2, shown in bold context above, are what this activity adds.)

R1 - Running Configuration

```
hostname R1
!
no ip domain-lookup
enable secret class
!
interface GigabitEthernet0/0
 description LAN to PC1
 ip address 172.16.1.1 255.255.255.0
 no shutdown
!
interface Serial0/0/0
 description Link to R2
 ip address 172.16.3.1 255.255.255.252
 no shutdown
!
interface Serial0/0/1
 description Link to R3
 ip address 192.168.10.5 255.255.255.252
 no shutdown
!
router ospf 1
 router-id 1.1.1.1
 passive-interface GigabitEthernet0/0
 network 172.16.1.0 0.0.0.255 area 0
 network 172.16.3.0 0.0.0.3 area 0
 network 192.168.10.4 0.0.0.3 area 0
!
line con 0
 password cisco
 logging synchronous
 login
line vty 0 4
```

```
 password cisco
 login
 transport input ssh telnet
!
end
```

R2 - Running Configuration

```
hostname R2
!
no ip domain-lookup
enable secret class
!
interface GigabitEthernet0/0
 description LAN to PC2
 ip address 172.16.2.1 255.255.255.0
 no shutdown
!
interface Serial0/0/0
 description Link to R1
 ip address 172.16.3.2 255.255.255.252
 no shutdown
!
interface Serial0/0/1
 description Link to R3
 ip address 192.168.10.9 255.255.255.252
 no shutdown
!
interface Serial0/1/0
 description Link to Internet/ISP
 ip address 209.165.200.225 255.255.255.224
 no shutdown
!
router ospf 1
 router-id 2.2.2.2
 passive-interface GigabitEthernet0/0
 network 172.16.2.0 0.0.0.255 area 0
 network 172.16.3.0 0.0.0.3 area 0
 network 192.168.10.8 0.0.0.3 area 0
 default-information originate
!
ip route 0.0.0.0 0.0.0.0 Serial0/1/0
!
line con 0
 password cisco
 logging synchronous
 login
line vty 0 4
 password cisco
 login
 transport input ssh telnet
```

```
!
end
```

R3 - Running Configuration

```
hostname R3
!
no ip domain-lookup
enable secret class
!
interface GigabitEthernet0/0
 description LAN to PC3
 ip address 192.168.1.1 255.255.255.0
 no shutdown
!
interface Serial0/0/0
 description Link to R1
 ip address 192.168.10.6 255.255.255.252
 no shutdown
!
interface Serial0/0/1
 description Link to R2
 ip address 192.168.10.10 255.255.255.252
 no shutdown
!
router ospf 1
 router-id 3.3.3.3
 passive-interface GigabitEthernet0/0
 network 192.168.1.0 0.0.0.255 area 0
 network 192.168.10.4 0.0.0.3 area 0
 network 192.168.10.8 0.0.0.3 area 0
!
line con 0
 password cisco
 logging synchronous
 login
line vty 0 4
 password cisco
 login
 transport input ssh telnet
!
end
```